

Synthetix Alpha

An autonomous, risk-gated trading agent on Alpaca

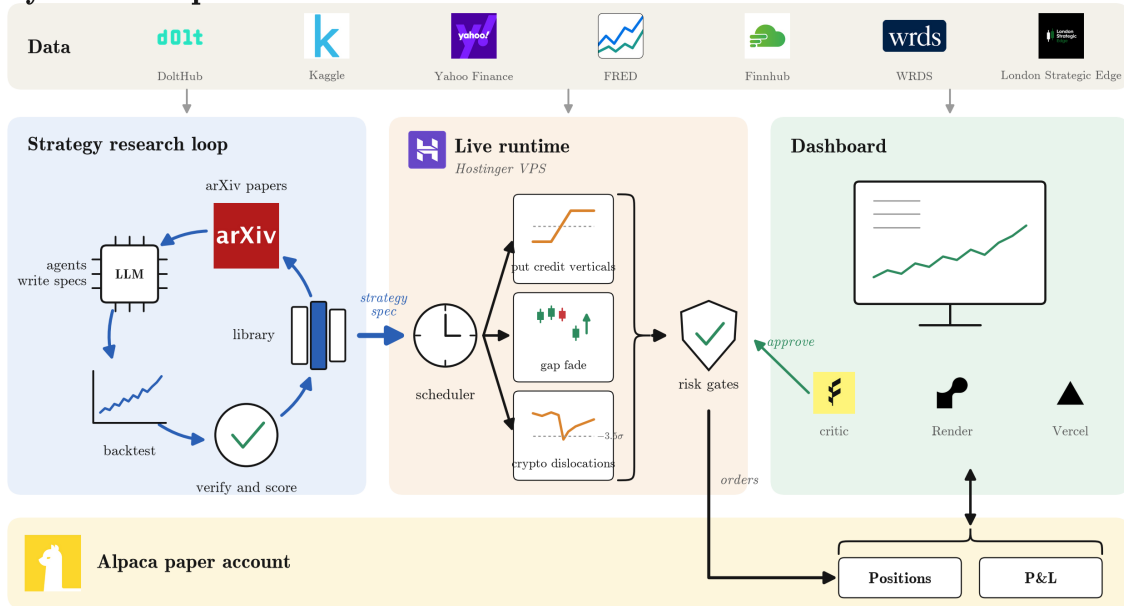
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Alpaca AI Trading Agents Hackathon · September 2026



Synthetix Alpha



One hard rule

LLM agents research. A deterministic runtime trades.

Where an LLM helps

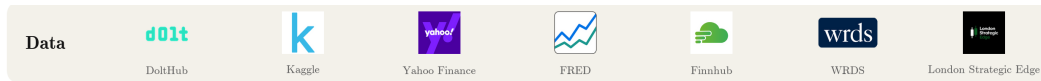
- reading quant-finance papers and turning their ideas into testable strategies
- writing strategy specs, not code, so every idea runs through the same backtester
- reviewing a candidate trade the way a risk officer would

Where it must not be

- the order path: every session runs the same rules, on a schedule, with no model call
- sizing and risk: fixed limits read from a config file, not a prompt
- the broker: the LLM never holds a key and never sees the CLI

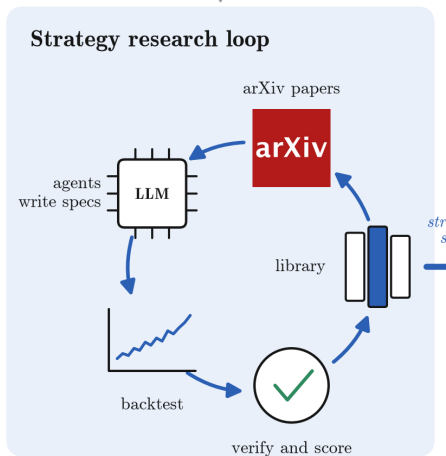
Live behaviour is reproducible line by line. The research is where the model earns its place.

Data



source	what it provides	used for
DoltHub	IV surfaces and HV/IV history, 1,500 names, 2019–2026	live screen, out-of-sample backtests
Kaggle	end-of-day option chains with greeks, 2016–2023	in-sample backtests
Yahoo Finance	earnings dates, split history	earnings gate, split adjustment
FRED	VIX, VXN, VXV, NFCI, macro series	volatility features, critic context
Finnhub	news, analyst and insider sentiment	critic context
WRDS	OptionMetrics and CRSP	spread and factor validation
London Strategic Edge	option chains, session bars	feed reconciliation
arXiv	q-fin papers	the research loop
Alpaca	chains, bars to one minute, crypto; paper trading	everything live

Strategy research loop



- **arXiv intake.** q-fin categories on a paced API, keyword relevance, PDFs into an append-only library.
- **Agents write specs.** One agent per paper writes candidate strategies in a JSON DSL. The brief carries what is already known, forbids tuning on the backtest, and treats gains under 0.54 Sharpe as noise.
- **Backtest.** Real chains replayed day by day.
- **Verify and score.** Robustness first; unseen vendor and unseen underlying; fragility sweep.
- **Library.** Append-only progress log; promotion to live is a human step.

140+ candidates evaluated; 29 papers in the library.

The spec DSL and the backtest engine

A strategy is data, not code

```
{ "name": "put_vertical_ivrv",  
  "legs": [  
    {"type": "put", "side": "short",  
      "delta": 0.20},  
    {"type": "put", "side": "long",  
      "delta": 0.10} ],  
  "underlyings": ["SPY", "QQQ"],  
  "dte_target": 65,  
  "dte_min": 40, "dte_max": 90,  
  "signal": {  
    "iv_rv_ratio": [1.27, null],  
    "vix_rv_ratio": [1.40, null] },  
  "profit_target": 0.65,  
  "stop_loss": 2.0, "dte_exit": 21,  
  "risk_fraction": 0.03,  
  "sizing": "max_loss",  
  "min_volume": 25, "min_credit": 0.05 }
```

What the engine does with it

- picks legs by delta, moneyness or width on the expiry nearest the DTE target
- fills at mid $\pm$ half the spread, \$0.65 per contract and leg; marks at mid; settles at intrinsic
- refuses contracts with no same-day volume: a mid in a contract that never traded is a quote, not a fill
- carries positions through stock splits the way the OCC does
- sizes by max loss on the payoff grid; exits on profit target, stop, DTE or hold limit
- 22 trailing-only features: IV surface, realised vol, VIX family, earnings, FOMC, technicals

Verify and score

$$\text{score} = \frac{1}{2} \bar{S} + \frac{1}{2} S_{\min} + 2 \cdot \text{worst year} + 3 \cdot \max(\text{maxDD}, -1) + (\text{positive years} - 1) \\ -9 \text{ under 40 trades, so a thin sample can never win}$$

- **Unseen vendor.** Fitted on Kaggle chains, re-run on DoltHub surfaces 2019–2026.
- **Unseen underlying.** AAPL was never used to fit the index rule.
- **Fragility sweep.** Every parameter perturbed one at a time; the median perturbed score is part of the objective from generation 3 on.
- **Concentration.** The top five trades must not carry the P&L.

put credit vertical	Sharpe	trades
in sample, Kaggle SPY+QQQ	0.92	102
DoltHub SPY 2019–2026	0.65	148
AAPL 2016–2023, unseen	0.49	184

The number we plan around is the out-of-sample one. The in-sample gap closed once the liquidity floor was added: the filter removed illusion, not edge.

Strategy evolution: the search

gen	strategy	Sharpe	score	what changed
0	put spread, IV-rank gate	-0.13	-1.18	hand-written baseline
0	put spread, IV/RV gate	0.73	0.40	conditional premium selling
0	diagonal put credit	0.85	0.75	generation winner, only 49 trades
2	put spread, IV/RV gate	1.01	0.98	winner, but ± 10 DTE collapses it to 0.28 / 0.07
3	put vertical, 65 DTE	1.14	1.09	re-centred onto the 55-70 DTE plateau; fragility fixed
4	put vertical, double gate	1.15	1.11	VIX or VXN must confirm the chain-derived gate
4	same, liquidity floor	0.92	0.52	25 contracts of same-day volume; 9.8% of earlier fills had none
5	single-name vertical, AAPL	0.90	0.84	earnings filter: no announcement within 30 days
7-8	three-sleeve portfolio	0.97	0.91	index, single-name and bear-call sleeves, equal weight

- 8 strategy angles, 2 seeds each; three generations of mutation and crossover on the top five; agents ran concurrently and each verified its own candidate.
- Corrections are logged even when the score falls. A number no longer believed does not stay the headline.

Strategy evolution: what survived

Kept

- selling put spreads only when implied vol is rich to realised; unconditional selling loses to costs
- 65 DTE, on the plateau rather than the edge of the 35–45 DTE hole; 20/10-delta wings; 65% profit target; exit at 21 DTE
- an earnings blackout for single names: AAPL score $-0.15 \rightarrow +0.84$
- equal weights across sleeves: every optimised weighting won in sample and lost out of sample

Tried and dropped

- long-vol structures: straddles, strangles and calendars carry negatively
- 60-day condors; 4-DTE technical-signal spreads (AAPL 0.80, QQQ -0.14)
- “skip the crisis”: dropping high-VIX days cut the score $1.09 \rightarrow 0.76$; panic is when the premium is richest
- the diagonal, once the vertical moved to 65 DTE; five paper ideas and a FOMC event rule, each rejected out of sample

Checked on OptionMetrics with overlap-corrected errors, the gate's advantage sits inside the noise. It stays as a filter; the expectation we carry is the out-of-sample Sharpe.

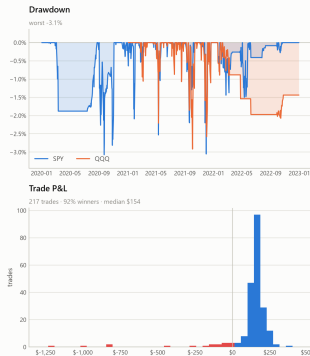
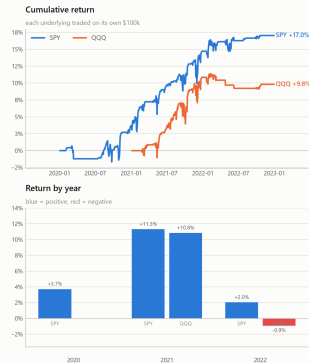
The options sleeve

- sell the 20-delta put, buy the 10-delta put, about 65 days out
- enter when $IV/RV \geq 1.27$ and $VIX/RV \geq 1.4$
- exit at 65% of the credit, a $2\times$ stop, or 21 DTE
- 3% of equity at risk per position, 12 slots, daily entry
- 25 contracts of same-day volume; credit at least 5% of max loss

Backtest 2020–2022 on SPY and QQQ
QQQ: 102 trades, Sharpe 0.92, max drawdown 2.0%.

put_vertical_ivrv — SPY, QQQ

Best Sharpe 1.05 · worst year -0.9% · max drawdown -3.1% · 217 trades



An intraday sleeve, and what minute bars showed

The gap fade

- 198 large caps; rank the overnight gap by 20-day volatility; buy the ten most negative at the open, flat by the close
- daily-bar backtest, five years: 4.6 bp a day net of measured spreads, Sharpe 0.65
- defects found and masked on the way: opens copied from closes, a busted opening auction, unadjusted splits, spin-off mismatches between feeds

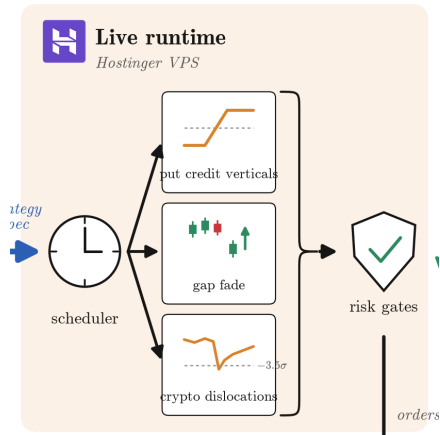
The loop re-measured its own strategy from the price an order actually meets and reported the answer.

Re-measured from the fill

leg, Alpaca 1-min bars	bp/day	<i>t</i>
official open → 09:31 print	+5.0	7.6
09:31 → 10:00	+0.1	0.1
10:00 → 15:50	+1.1	0.4
select on an independent open	+0.3	0.5

The whole backtested edge sits between the auction print and the first trade after it, a price no market order can get. Selecting on a different noisy open removes it.

Live runtime



- **Scheduler** on a Hostinger VPS: 09:31 enter, 09:45 top-up, 15:50 and 15:56 flatten (ET). State is marked before firing, so a crash never retries; a cron watchdog restarts it within five minutes.
- **A clean subprocess per fire**, with its own credentials, dry run unless told to execute.
- **Three sleeves**: put credit verticals; the equity gap fade at 150% of NAV, flat by the close; crypto dislocations, a 24-hour move below -3.5σ held 48 hours, at most three open.
- **Windows**: entries 09:31–10:30, flattening 15:30–15:58, weekends refused.
- **Alpaca CLI** for every account read and order; the SDK only for market data.

Risk gates and execution

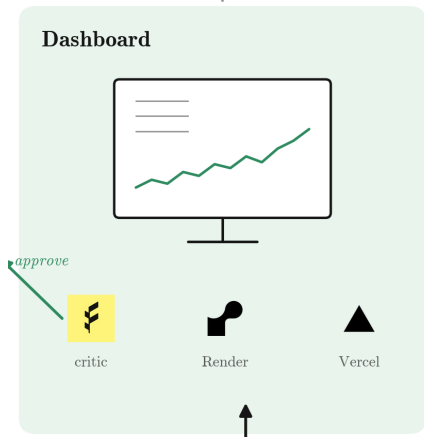
Before an order exists (governance.yaml)

- NAV must be positive; halt at 20% total or 5% daily drawdown
- leverage 1×; 10% of NAV per name; 12 open positions
- defined-risk structures only; 3% of equity at risk per position
- screen: IV/RV in 1.25–2, \$20M average daily volume, chain spread under 10%, no earnings within 30 days
- crossing check: the credit at bid and ask must clear 60% of the mid credit
- corporate-action guard: a large cap does not open 25% lower on anything a fade can trade

Once it does

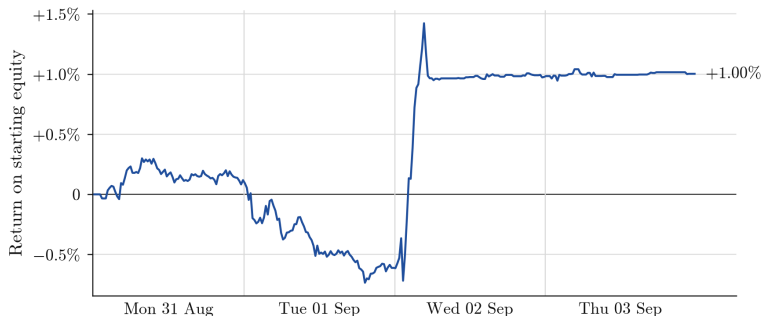
- multi-leg limit orders, up to four legs, day time in force
- client order id = hash of the legs and the date; the same spread on the same day is the same id, checked in a local ledger and against the broker's open, closed and all orders
- status is never “filled” unless Alpaca said so
- exits sized to the actual fill; flattening covers only what is held and skips anything resting
- a paper-only assertion refuses to run if live trading is switched on anywhere in the environment

Operator dashboard and the Featherless critic



- **FastAPI** adapter in a Docker image with the Alpaca CLI, on Render; **Next.js** command centre on Vercel: pipeline trace, opportunities, positions, execution ledger, research view.
- **Pipeline:** screen → gather (Finnhub, FRED) → critique → form → risk → operator approval → paper execution.
- **The critic** is a Featherless-hosted model prompted as a risk officer: approve or reject with a confidence and a size multiplier; reject on imminent earnings, a deep curve inversion with tightening conditions, or adverse headlines.
- **Advisory by construction:** every approval re-runs the deterministic risk gate on that single order, and the operator confirms in two steps.

Paper trading, 31 August to 3 September



- four sessions, 5-minute marks, regular hours
- every scheduled fire on time: entries, top-ups, flattens
- an options spread filled and carried through the week
- every gate, order id and fill reconciled against the broker

What we would do next

- **Rank on pre-market quotes and buy the auction.** The only place the gap fade's edge exists is the opening print itself, which needs market-on-open orders and a ranking that never sees the print.
- **Put the critic in the loop with a live key** and measure its approvals against realised outcomes, the way every other component was measured.
- **Larger option capacity.** The rule runs about one contract a trade; the same Sharpe scales to twenty times the return with daily entry and more slots.
- **Keep the research loop running.** It is append-only, it logs its corrections, and it has already caught its own best-looking result.

github.com/alan-d-smith/synthetix-alpha